Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 7 · day per timetable	7	1 period (60 min)	Completing the Square	Algebra II	Topic 2 — Quadratic Functions and Equations · Lesson 2-5	Grade 10 — 10A and 10C

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · Whiteboards · Canva (final product) · Squared paper	Lesson presentation (15 slides) · "Choose Your Route" differentiation sheet · "Corniche Water Jets" project task (A3) · "Give it a go" classwork

Competencies	Technology Embedded	Connection to student's real life
1- Finding the adjustment term and solving quadratic equations by completing the square, including complex solutions. 2- Finding the minimum or maximum value of a quadratic function and where it occurs.	1- Quizizz live diagnostic with instant gap analysis. 2- Calculator check: confirm a solution by substitution and round only at the last step.	1- A model water jet: the greatest height, the time it happens and the time to reach the sea. 2- A rectangle with a fixed perimeter and its greatest area.

Values	Soft Skills	Hard Skills
1- Balance — what is done to one side is done to the other. 2- Precision — rounding only at the last step.	1- Collaboration in assigned group roles. 2- Explaining in words why the adjustment term is added to both sides.	1- Completing the square and rewriting an equation as $(x-p)^2=q$. 2- Finding extreme values of a quadratic function.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Rewrite a quadratic equation into the form $(x - p)^2 = q$ by completing the square. 2- Determine the minimum or maximum value of a quadratic function using completing the square. 3- Solve real-world problems modeled by quadratic equations using completing the square. *Vocabulary: adjustment term ; completing the square.*	FIKR Phase 1 — Diagnose. 3-question Quizizz: $(x+4)^2$, $x^2+10x+25$, $(x-3)^2=16$. Results read as a gap map; students routed to the re-teach table, straight to the adjustment term, or the Investigate prompt. FIKR Phase 2 — Targeted Instruction. Whole class: the adjustment term; solving by completing the square; $a \neq 1$; maximum and minimum values. Quick check on whiteboards. FIKR Phase 3 — Practice. Two identical "Give it a go" problems worked together. Students then choose a route and may switch at any time. Live feedback via Pear Deck; 2-minute micro-conferences. FIKR Phase 4 — Production. The Corniche water jet: vertex form, the greatest height, two equal-height times and the time to sea level. AI used as critic only. Extended in the A3 project task. FIKR Phase 5 — Proof of Learning. "Time to Check" — an adjustment term, an equation and a minimum value. No notes, no partner, no AI. FIKR Phase 6 — Smart Production. Students refine their reasoning into a final product including one worked non-example, and publish to the LMS portfolio.	Diagnostic quiz — gap map, not a grade. Quick check for understanding on whiteboards (80% threshold). "Give it a go" questions — self-marked, teacher circulates. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. PASS → Enrichment & Challenge. NOT YET → Targeted Learning Clinic next lesson. Final product graded on reasoning and clarity, not correctness alone.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Two minutes of peer show-and-tell on the final products, then the reflection question: "Which is harder for you — finding the adjustment term, or keeping the equation balanced?"
Reflection for Improvement:	
Homework:	Finish the final product and upload to the LMS portfolio (any of the three formats). Practice set on Pear Deck — 12 questions, choose your route. Watch the flipped video for the next lesson, Topic 2 · Lesson 2-6 — The Quadratic Formula. Clinic group: bring your Time to Check paper.

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